Use Cases for Mist:

Recommended Use Cases for Each Mist Script

MistTrackerVulkan.js

- **Primary Use Case:**
  Core data engine for tracking, CRUD, session/state management, and advanced navigation in 3D/4D/nD.
- **Standalone/Subset Use Case:**
  - Single-user timeline and relationship tracker.
  - 3D/4D data visualization and navigation tool.
  - Data modeling and provenance tracking for research or creative projects.

MistIllum.js

- **Primary Use Case:**
  Advanced rendering, physics, audio, and UI for immersive and physically-based visualization.
- **Standalone/Subset Use Case:**
  - Physics-based 3D/4D renderer for scientific or educational demos.
  - Audio/visualization module for other engines.
  - Standalone wave/interference simulation tool.

MistMulti.js

- **Primary Use Case:**
  Real-time multi-user, P2P, and moderation/collaboration layer.
- **Standalone/Subset Use Case:**
  - Secure P2P session manager for collaborative apps.
  - Distributed moderation and provenance system.
  - Real-time presence and event sync for other collaborative tools.

pureMathPhysicsEngine.js

- **Primary Use Case:**
  Mathematical and physics foundation for all metric, wave, and probability logic.
- **Standalone/Subset Use Case:**
  - Physics/math library for scientific computing.
  - Standalone metric tensor and wave simulation toolkit.
  - Probability and anomaly detection module for other systems.

Recommended Use Cases for Subsets (Excluding All Combined)

- MistTrackerVulkan.js **+ MistIllum.js:**
  Single-user, high-dimensional visualization and simulation tool for research, education, or creative world-building.
- MistTrackerVulkan.js **+ MistMulti.js:**
  Collaborative data tracker for timelines, events, and relationships (e.g., collaborative story analysis, project management).

- MistIllum.js **+ MistMulti.js:**
  Real-time collaborative physics/visualization sandbox for education or remote labs.
- MistIllum.js **+ pureMathPhysicsEngine.js:**
  Standalone scientific visualization and simulation platform with advanced math/physics.
- MistMulti.js **+ pureMathPhysicsEngine.js:**
  Distributed anomaly detection and moderation system for collaborative scientific or creative projects.

Note:
The primary use case of all scripts combined is a collaborative, high-dimensional, physically-based world/narrative tracker and simulator with advanced rendering, audio, and moderation. The above recommendations focus on alternative or specialized uses for each script or subset.